

KUMLINGE ISLAND, Finland

WMO: 029900

Lat: 60.300N

Lon: 20.750E

Elev: 5

StdP: 101.26

Time Zone: 2.00 (EUE)

Period: 86-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.0	-13.6	-19.0	0.7	-16.6	-15.6	1.0	-13.6	19.2	1.9	16.7	2.9	5.1	30	0.815

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.3	22.8	18.1	21.2	17.2	19.7	16.4	18.7	21.8	17.7	20.4	16.8	19.2	4.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	12.5	20.4	16.5	11.7	19.4	15.6	11.1	18.4	53.1	22.0	49.8	20.4	47.1	19.4	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.6	15.1	13.4	DB	-14.9	25.4	7.9	1.6	-20.6	26.6	-25.2	27.5	-29.7	28.4	-35.4	29.6	(4)
(5)			WB	-15.4	19.9	7.2	1.2	-20.6	20.8	-24.8	21.5	-28.8	22.2	-34.0	23.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-1.9	-3.1	-0.8	3.1	7.8	12.8	15.9	15.4	11.7	7.4	3.2	-0.2	(6)
(7)		DBStd	7.36	5.21	5.25	3.88	2.82	2.88	2.81	2.31	2.40	2.54	2.68	3.00	3.27	(7)
(8)		HDD10.0	1974	368	366	333	206	81	6	0	0	11	84	203	317	(8)
(9)		HDD18.3	4512	626	599	592	456	326	168	83	96	199	339	453	575	(9)
(10)		CDD10.0	518	0	0	0	0	13	89	182	168	62	4	0	0	(10)
(11)		CDD18.3	14	0	0	0	0	0	2	7	6	0	0	0	0	(11)
(12)		CDH23.3	31	0	0	0	0	0	3	19	9	0	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	6.6	8.0	7.0	6.3	6.2	5.5	5.3	5.6	5.9	6.4	7.9	7.2	7.9	(14)
(15)	Precipitation	PrecAvg	835	87	64	53	42	43	61	74	79	70	83	86	94	(15)
(16)		PrecMax	1040	147	136	98	71	91	112	138	144	142	170	197	165	(16)
(17)		PrecMin	618	13	2	10	6	10	25	4	27	20	12	13	18	(17)
(18)		PrecStd	101	37	33	21	18	21	22	33	30	37	38	37	37	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	5.3	7.2	13.8	19.8	23.6	25.7	24.5	20.0	13.1	9.7	6.3	(19)
(20)			MCWB	3.8	4.1	4.9	9.7	14.5	18.2	19.7	19.0	16.6	11.8	8.6	5.2	(20)
(21)		2%	DB	4.0	4.1	5.2	11.4	16.6	21.2	23.1	22.2	18.1	12.1	8.8	5.3	(21)
(22)			MCWB	2.8	3.2	3.6	8.2	12.3	16.3	18.1	18.2	15.2	11.1	7.9	4.5	(22)
(23)		5%	DB	3.3	3.0	4.1	9.4	14.6	19.2	21.6	20.5	16.5	11.5	8.0	4.7	(23)
(24)			MCWB	2.4	2.4	2.9	7.0	10.6	15.0	17.5	17.4	14.2	10.5	7.1	4.0	(24)
(25)		10%	DB	2.6	2.3	3.1	7.5	12.7	17.4	20.0	19.1	15.2	10.8	7.1	4.0	(25)
(26)			MCWB	1.9	1.7	2.2	5.6	9.5	14.1	16.7	16.5	13.2	9.8	6.3	3.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.5	5.2	10.0	15.0	19.0	20.6	20.0	17.4	12.2	8.9	5.6	(27)
(28)			MCD B	4.7	5.1	6.4	13.3	19.5	23.1	23.7	23.3	19.4	12.7	9.4	6.0	(28)
(29)		2%	WB	3.2	3.4	3.9	8.5	12.6	17.1	18.8	18.7	15.7	11.3	8.1	4.8	(29)
(30)			MCD B	3.6	3.8	4.8	11.1	16.5	20.0	21.9	21.6	17.6	11.9	8.7	5.2	(30)
(31)		5%	WB	2.6	2.6	3.1	7.1	10.9	15.5	17.8	17.7	14.5	10.7	7.2	4.2	(31)
(32)			MCD B	3.1	2.9	3.9	9.3	14.0	18.2	20.9	19.9	16.1	11.3	7.8	4.6	(32)
(33)		10%	WB	2.0	1.9	2.3	5.7	9.7	14.4	16.8	16.8	13.6	10.0	6.4	3.4	(33)
(34)			MCD B	2.5	2.2	3.0	7.3	12.3	17.1	19.7	18.7	14.9	10.6	7.0	3.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.3	4.1	3.9	4.8	5.9	5.3	4.3	3.9	2.9	2.9	3.2	(35)	
(36)			MCD BR	3.2	3.2	4.7	8.3	9.5	8.8	8.2	6.7	5.3	3.6	2.8	3.0	(36)
(37)			MCWBR	3.0	2.8	3.4	5.6	5.7	4.9	4.1	3.3	3.4	3.2	2.8	3.0	(37)
(38)		5% WB	MCD BR	2.9	2.9	4.2	8.2	9.0	7.7	7.4	5.9	4.9	3.5	2.8	3.0	(38)
(39)			MCWBR	3.0	2.8	3.3	5.7	5.6	4.7	4.0	3.2	3.4	3.3	3.0	3.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.311	0.330	0.369	0.367	0.358	0.387	0.376	0.347	0.333	0.300	0.385	(40)	
(41)		taud	2.175	2.248	2.300	2.324	2.421	2.490	2.435	2.451	2.514	2.411	2.217	2.639	(41)	
(42)		Ebn at Noon	483	678	795	827	860	874	835	815	781	656	441	338	(42)	
(43)		Edh at Noon	41	68	89	105	102	97	101	92	73	59	40	29	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.23	0.84	2.39	4.03	5.53	6.10	5.67	4.41	2.69	1.14	0.31	0.14	(44)	
(45)		RadStd	0.03	0.15	0.27	0.38	0.47	0.41	0.51	0.38	0.30	0.17	0.06	0.02	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (44 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	-114	N/A	N/A	N/A	

Nomenclature: See separate page